

FLYN AI — Contracts & eSignature Engine

Engineering Implementation Document (Internal)

1. Objective

Implement a cross-module Contracts & eSignature Engine to generate, send, sign, and store agreements.

This module is shared across CRM, HR, Freelance, Website Builder, and API layers.

2. Module Placement

Core Module: Modules/Contracts/

Must not be embedded inside any single feature module.

Exposed as internal service + public API.

3. Folder Structure

Modules/Contracts/

- Controllers (ContractController, SigningController)
- Services (ContractService, SignatureService, RenderService)
- Models (Contract, Signature, Template, Signer)
- Templates (HTML/PDF templates)
- Routes (api.php)

4. Core Features

- Contract generation from templates (NDA, Employment, Freelance, Sales)
- AI contract generation (dynamic clauses and fields)
- Digital signatures (draw, type, upload)
- Shareable signing links with expiry and one-time tokens
- Multi-signer workflows (client, freelancer, employer)
- Contract status tracking (draft, sent, viewed, signed, declined, expired)
- PDF generation and locking after signing
- Audit trail (timestamp, IP, user agent)
- Storage and versioning

5. Data Model (Core Tables)

contracts:

- id, organization_id, template_id, status, file_url, created_at

signers:

- id, contract_id, name, email/phone, role, order, status

signatures:

- id, contract_id, signer_id, signature_data, signed_at, ip_address

contract_templates:

- id, organization_id, name, html_template, variables JSON

contract_events:

- id, contract_id, type, payload JSON, created_at

6. Signing Flow

1. Contract created (template or AI)
2. Signers assigned
3. Unique secure link generated
4. Contract sent via WhatsApp/Email/Link
5. Recipient opens link and signs
6. Signature stored and validated
7. Final PDF generated and locked
8. Status updated to 'signed'
9. Triggers automation (payment, onboarding)

7. API Endpoints

POST /api/contracts

GET /api/contracts

GET /api/contracts/{id}

POST /api/contracts/{id}/send

POST /api/contracts/{id}/sign

POST /api/contracts/{id}/void

GET /api/contracts/{id}/download

GET /api/contracts/{id}/events

8. Service Layer

ContractService::create()

ContractService::send()

ContractService::complete()

SignatureService::capture()

RenderService::generatePDF()

9. Integration Points

CRM: deal → contract generation

HR: onboarding → employment contracts

Freelance: job accepted → freelancer agreement

Website Builder: form submission → contract trigger

Automation Engine: contract_signed → next actions

10. Security & Compliance

- JWT-secured API access
- Signed URLs with expiry
- AES-256 encryption for sensitive data
- Audit logs for all actions
- IP + timestamp logging
- Optional integration with third-party providers (DocuSign, Adobe Sign)

11. Storage & Delivery

- Store signed PDFs in secure storage (S3 or equivalent)
- Provide downloadable links
- Maintain version history
- CDN delivery for performance

12. Automation Hooks

contract.created

contract.sent

contract.viewed

contract.signed

contract.declined

Used to trigger CRM, HR, or payment workflows

13. Implementation Roadmap

Phase 1: Contract creation + templates + PDF rendering

Phase 2: Digital signature capture + signing links

Phase 3: Multi-signer + workflow automation

Phase 4: AI contract generation + analytics

Phase 5: Third-party integrations + compliance enhancements

14. Final Outcome

A centralized legal agreement engine powering all FLYN modules.

Enables end-to-end workflows from deal → contract → payment → automation.